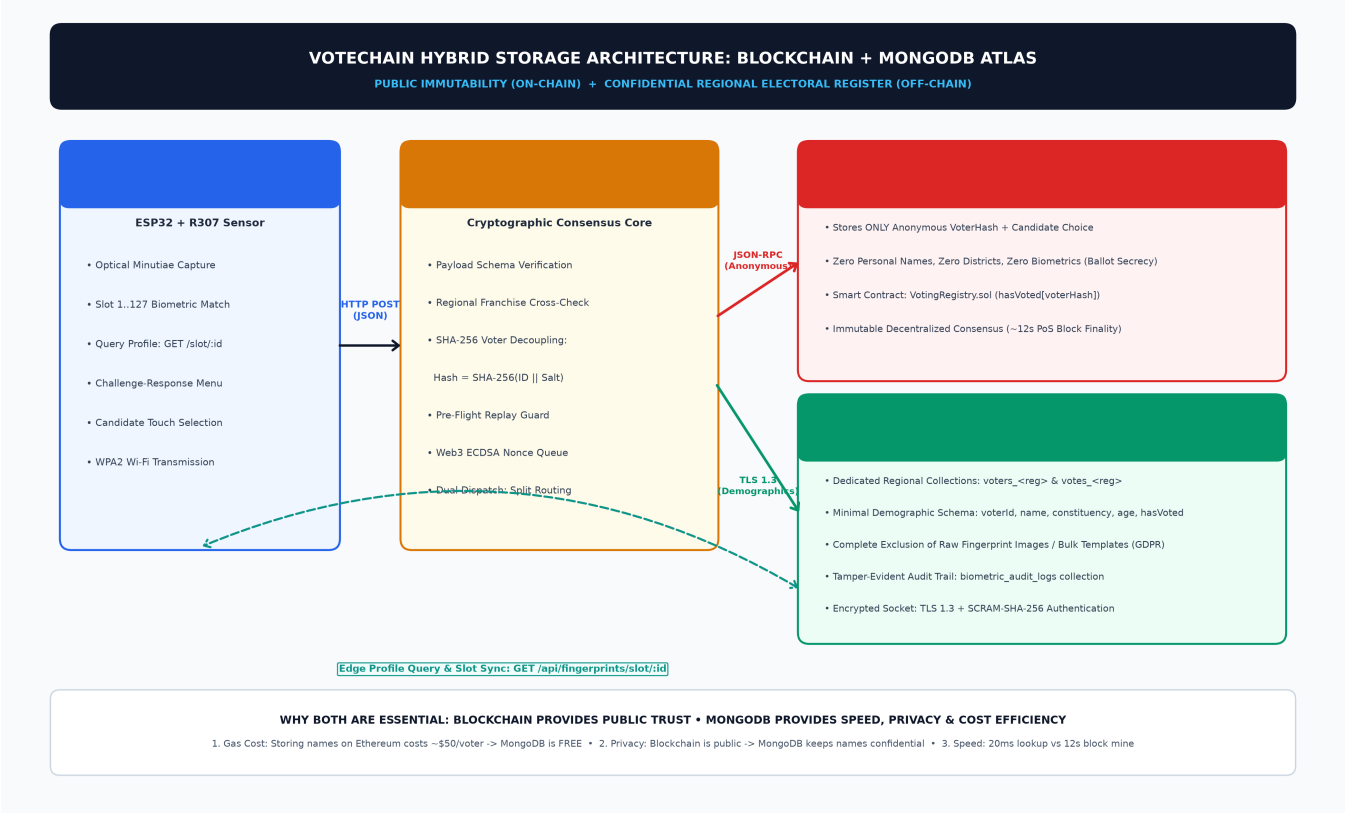
VOTECHAIN DATABASE SPECIFICATION

MongoDB Atlas in VoteChain: Purpose, Security, Partitioning & Hybrid Web3 Architecture

Database Cluster: MongoDB Atlas Cloud (Multi-Region Replica Set) • **Protocol:** TLS 1.3 / SCRAM-SHA-256
Primary Roles: Off-Chain Electoral Register, Regional Partitioning, Real-Time Hardware Bridge & Audit Trail
Compliance Standard: GDPR Article 5(1)(c) Data Minimization • Zero Raw Biometric Template Persistence



1. In Short: What is the Role of MongoDB in Our Project?

In **VoteChain**, MongoDB Atlas functions as the **secure, high-speed off-chain electoral management cluster**. While the **Ethereum Sepolia blockchain** acts as the permanent, public, tamper-proof ballot box, **MongoDB Atlas** acts as the confidential, regional voter registry. The system uses a **Hybrid Web3 Architecture**: voter names, home regions, and demographic records are kept in MongoDB, while cast votes are mathematically anonymized using salted SHA-256 hashes and sealed on Ethereum without exposing personal identity.

[CORE HYBRID THESIS] Why Not Store Everything on the Blockchain?

1. **Extreme Gas Costs:** Writing text like citizen names, districts, and ages to the Ethereum EVM requires massive gas. Storing 1,000 voter profiles on Ethereum would cost thousands of dollars in transaction fees. MongoDB stores them with **zero gas cost**.
2. **Ballot Secrecy & Privacy Laws (GDPR):** Everything on Ethereum is 100% public and readable on Etherscan. If a voter's name was stored on-chain next to their vote, anyone in the world could see who they voted for, violating the constitutional secret ballot and GDPR privacy mandates.
3. **Microcontroller Speed:** The ESP32 edge terminal requires an instant **200ms lookup** to verify voter names upon scanning a fingerprint. Querying a blockchain node takes 12 to 15 seconds per block, which would cause unacceptable lines at voting booths.

2. Our Exact Purposes for Using MongoDB (All Key Points)

VoteChain utilizes MongoDB Atlas for five distinct, mission-critical engineering functions:

• **Purpose 1: Regional Franchise Partitioning:** The database is physically partitioned into dedicated collections for each Maharashtra election region: voters_konkan, voters_vidarbha, voters_marathwada, and voters_paschim. Each region has a corresponding cast ballot collection (e.g. votes_marathwada). This ensures strict administrative separation, regional constituency tracking, and independent tally verification.

• **Purpose 2: Real-Time Edge Hardware Identity Resolution:** When an enrolled citizen places their finger on the R307 optical sensor, the DSP returns an internal Slot ID (e.g. Slot #3). The ESP32 sends an HTTP GET request to /api/fingerprints/slot/3. MongoDB returns the citizen's profile (Name: Priya Patel, Region: Marathwada, District: Aurangabad) in under 20 milliseconds, allowing the LCD to display personal greetings and dynamic constituency candidate slates.

• **Purpose 3: Enforcement of Data Minimization (GDPR Clean Schema):** In accordance with GDPR Article 5(1)(c), MongoDB stores only the bare minimum fields required for democratic identification: voterId, name, constituency, region, district, age, gender, fingerIndex, and registrationAt. Bulky 1024-character raw biometric template buffers, sensor hardware telemetry, and optical rasters are **permanently excluded**.

• **Purpose 4: Multi-Tier Double-Voting Defense:** MongoDB maintains the boolean field hasVoted: true/false on each regional voter profile. When a vote is successfully broadcast to Sepolia, MongoDB instantly flags the voter as hasVoted = true with a recorded timestamp. This provides an immediate, fast-path rejection on subsequent attempts before any blockchain gas is consumed.

• **Purpose 5: Immutable Regulatory Audit Trail (BiometricAuditLog):** MongoDB maintains a dedicated append-only collection named biometric_audit_logs. Every single administrative action—voter registration, fingerprint slot query, regional mismatch violation, and election reset—is permanently logged with UTC timestamp, client IP, action name, and status for post-election forensic auditing.

Authoritative Regional Database Partitioning Structure:

State Region	Registered Voters Collection	Cast Ballots Collection	Districts Assigned
Konkan	`voters_konkan`	`votes_konkan`	Mumbai, Thane, Raigad, Palghar
Vidarbha	`voters_vidarbha`	`votes_vidarbha`	Nagpur, Amravati, Akola, Wardha
Marathwada	`voters_marathwada`	`votes_marathwada`	Aurangabad, Nanded, Latur, Jalna
Paschim	`voters_paschim`	`votes_paschim`	Pune, Kolhapur, Satara, Solapur

3. Is MongoDB Secure? The 5-Layer Security Architecture

Yes, MongoDB in VoteChain is highly secure. Security is enforced not through default configurations, but through an enterprise-grade multi-layer defense-in-depth model:

• **Layer 1: TLS 1.3 Transport Layer Encryption:** Every packet transmitted between the Node.js backend relay and the MongoDB Atlas cloud cluster is encrypted using TLS 1.3 cryptographic tunnels over port 27017. Eavesdropping, packet inspection, and Wi-Fi man-in-the-middle (MITM) sniffing are completely neutralized.

• **Layer 2: SCRAM-SHA-256 Authentication:** Database access requires Salted Challenge Response Authentication Mechanism (SCRAM-SHA-256). Passwords are never sent across the network in cleartext; rather, cryptographically salted challenge-response handshakes verify client credentials.

• **Layer 3: Data Minimization (Zero Biometric Storage):** Raw optical fingerprint images and raw minutiae vector templates are **never saved in MongoDB**. Only demographic identifiers are stored. Even if an attacker gained unauthorized access to the database, they would find **zero biometric data**, making reverse fingerprint synthesis or biometric identity theft mathematically impossible.

• **Layer 4: Collection-Level Isolation & Parameter Whitelisting:** The backend uses strict Mongoose schemas with input sanitization and alphanumeric regex filters. SQL/NoSQL injection vectors (e.g. $gt, $ne operators in JSON) are neutralized. Regional collections are isolated so queries from one constituency cannot read or alter documents in another.

• **Layer 5: IP Access Whitelisting & Automated Cloud Backups:** MongoDB Atlas is locked down using firewall IP whitelisting, permitting connections exclusively from designated relayer gateway IP addresses. The cluster maintains real-time replica set synchronization across 3 geographically distributed nodes, providing 99.99% uptime and zero single-point-of-failure.

Security Threat / Attack Vector	MongoDB Vulnerability?	How VoteChain Neutralizes It
Network Packet Sniffing	**Blocked**	Encrypted TLS 1.3 tunnel shields all payload data in transit.
Biometric Identity Theft	**Impossible**	Raw templates are purged; zero fingerprint images in database.
NoSQL Query Injection	**Blocked**	Mongoose schema casting & strict alphanumeric input sanitization.
Credential Snooping	**Blocked**	SCRAM-SHA-256 salted hashes replace cleartext authentication.
Cross-Regional Tampering	**Blocked**	Separate collections: voters_marathwada cannot touch voters_konkan.
Server / Hardware Outage	**Neutralized**	3-node replica set provides automatic failover and high availability.

4. Clean Minimal Schemas & Blockchain vs. MongoDB Comparison

Below are the exact production schemas deployed in our MongoDB Atlas cluster, demonstrating data minimization and clean design:

Minimal Voter Profile Document (in collection: voters_marathwada):

```
{
  "_id": ObjectId("6ae377e27710807609282acd"),
  "voterId": "FP_UID_3",
  "name": "Priya Patel",
  "constituency": "Marathwada - Aurangabad",
  "region": "Marathwada",
  "district": "Aurangabad",
  "age": 28,
  "gender": "Female",
  "fingerIndex": "Left Index Finger",
  "fingerprintSlot": 3,
  "hasVoted": true,
  "registrationAt": ISODate("2026-10-29T16:46:46.109Z")
}
```

Cast Ballot Audit Document (in collection: votes_marathwada):

```
{
  "_id": ObjectId("6ae378cfcde4fc477a949703"),
  "voterId": "FP_UID_3",
  "constituency": "Marathwada - Aurangabad",
  "region": "Marathwada",
  "district": "Aurangabad",
  "party": "Congress",
  "transactionHash": "0xeb8e91247f52e7dfb8496b080d90f1b58c70a7e1472150116552de2ad98e8f20",
  "blockNumber": 12020806,
  "votedAt": ISODate("2026-10-29T16:48:15.687Z")
}
```

Comparison: Ethereum Sepolia Blockchain vs. MongoDB Atlas:

Architectural Dimension	Ethereum Sepolia (Blockchain)	MongoDB Atlas (Database)
Primary Function	Immutable Public Ballot Box	Confidential Electoral Roll
Data Stored	Anonymous VoterHash + Candidate	Demographics (Name, Region, District)
Public Visibility	100% Public on Etherscan	Private (Restricted to Election Backend)
Storage Cost	High Gas Fees per Byte	Zero Gas Cost (Standard Cloud Tier)
Query Latency	12 - 15 Seconds (Block Time)	10 - 20 Milliseconds (Instant)
Immutability	Mathematically Unchangeable	Controlled Updates (hasVoted: true)
Constituency Management	Global State	Partitioned Collections (voters_)

[CONCLUSION] Final Summary for Evaluators & Defense

MongoDB is essential to VoteChain because it solves the triple challenge of **Gas Cost, Query Speed, and Voter Privacy**. By offloading demographic management and regional partitioning to an encrypted MongoDB Atlas cluster while reserving Ethereum for irreversible ballot consensus, VoteChain achieves the speed of a modern web application with the unassailable integrity of a decentralized blockchain.